Scenario 4A: Sparse-count observation stress test with continuous-time x2

MSSTNB posterior performance, fit-status classification, and oracle diagnostics

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1 Purpose and main conclusion

Scenario 4A is a sparse-count observation stress test for the MSSTNB model under the same continuous-time covariate design used in Scenarios 1–3. The stress is introduced at the observation level by lowering the expected count intensity while preserving the reference latent temporal-spatial structure. The goal is not to make every replicate look stable. The goal is to identify what fails first when the observed counts become weakly informative.

The main result is:

> Under the continuous-time x_2 design, Scenario 4A preserves good regression-slope recovery but still produces latent-scale numerical instability in selected replicates. The dominant failure mode is not failure of the beta/r block. It is a sparse-count identifiability problem involving the intercept, spatial effects, and latent temporal risk λ.

The formal fixed-gamma sampled-λ analysis produced 6 stable fits out of 10 replicates and 4 numerical-instability fits. There were no soft-warning replicates. The numerical-instability replicates were 1, 4, 8, 10.

2 Scenario design

Scenario 4A uses the same dynamic latent structure as the reference dynamic scenarios but lowers the observation-level count intensity. The official settings for this continuous-time version are:

Table 1: Official Scenario 4A continuous-time x2 design.

Quantity	Value
Scenario ID	S4A_SPARSE_COUNTS_CONTINUOUS_X2_T100
Stress type	Observation-level sparse counts
Time points	100
Coarse areas	9
x2 mode	continuous_time
x2 AR coefficient	0.50
x2 innovation SD	0.80
Sparse intercept shift	-4.25
Reference intercept	-1.50
Sparse raw intercept	-5.75
Fixed gamma	0.80

The sparse observation shift is

$$\text{sparse\_beta0\_shift} = -4.25,$$

so the expected count multiplier is

$$\exp(-4.25) \approx 0.014.$$

This makes the observed counts sparse while keeping the latent temporal-spatial process available for posterior and oracle diagnostics.

3 Sparse-count data summary

The generated data satisfy the intended continuous-time x_2 design. The empirical lag-one correlation of x_2 is close to the target value of 0.50, and the covariate is not binary-like.

Table 2: Sparse-count data summary for S4A continuous-time x2.

Quantity	Value
Number of replicates	10
Average count	2.905
Average median count	1.700
Average zero proportion	0.249
Total count	26,149
Maximum observed count	61
x2 empirical AR(1), average	0.478
x2 binary-like proportion, max	0.000

The average count is 2.905 and the average zero proportion is 0.249. Thus, this is a sparse-count regime, but not an all-zero regime.

4 Fitting strategy

The main S4A fit uses the dynamic MSSTNB structure with fixed $\gamma = 0.8$. The model estimates

$$\beta_0,\ \beta_1,\ \beta_2,\ \phi,\ \tau_\phi,\ r,\ \kappa,\ \lambda,$$

while γ is fixed at its truth. The split component ω is disabled in this S4A analysis. Numerical guards are not hidden. They are recorded and used to classify each replicate.

The fit-status rule is:

- **Stable:** no beta, kappa, or lambda-output guard, and no extreme intercept or latent-risk metric.
- **Soft warning:** beta guard only, with no kappa or lambda-output instability.
- **Numerical instability:** kappa guard, lambda-output guard, extreme β_0, extreme stored $\log\lambda$, or extreme $\log\lambda$ recovery error.

5 Fit-status classification

Table 3: Fit-status classification for S4A continuous-time x2.

Status	# reps	Percent
Stable	6	60.0
Numerical	4	40.0

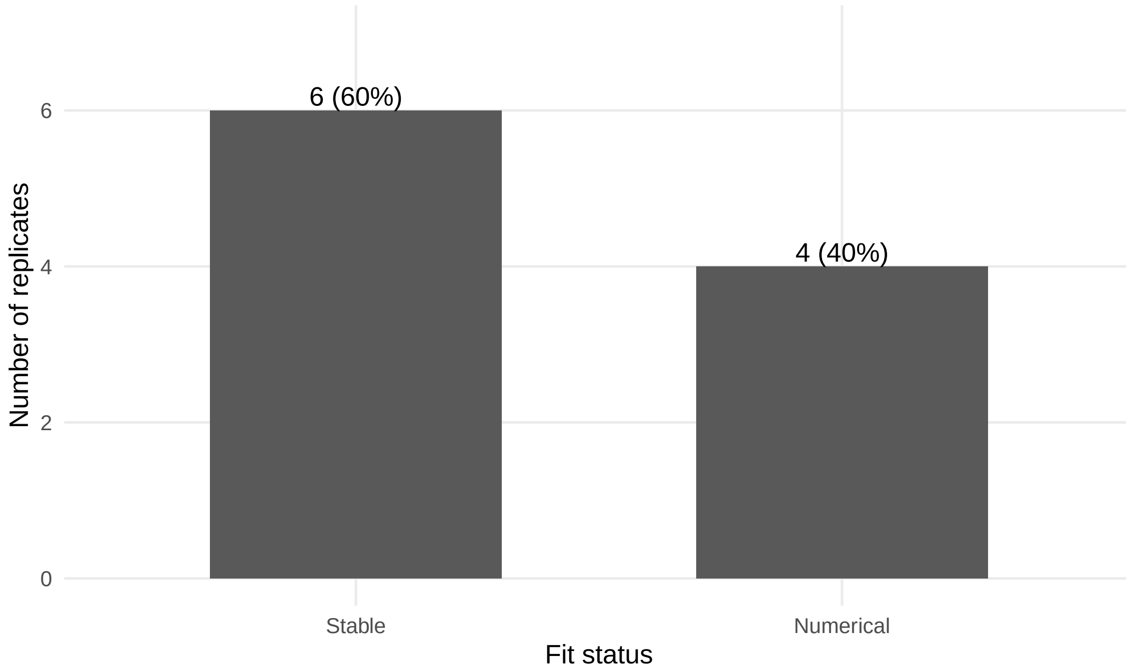

Figure 1: Fit-status counts for S4A continuous-time x2.

The numerical-instability rate is 40.0%. Importantly, this instability is not a regression-slope failure. As shown below, β_1 and β_2 remain close to truth even in the numerical-instability subset.

6 Posterior performance by subset

The full posterior-performance table is split into small tables to avoid hiding the main message in an unreadably wide table. Because there are no soft-warning replicates, the stable-plus-soft subset is identical to the stable subset and is omitted from the displayed subset tables.

Table 4: Observation intensity by fit-status subset.

Subset	# reps	Mean count	Zero prop.
All	10	2.905	0.249
Stable	6	3.191	0.214
Instab.	4	2.478	0.302

Table 5: Regression and dispersion recovery by fit-status subset.

Subset	# reps	b0 mean	b0 cover	b1 mean	b1 cover	b2 mean	b2 cover	r mean	r cover
All	10	-6.170	0.300	0.493	0.900	-0.385	1.000	16.513	1.000
Stable	6	-5.860	0.500	0.493	0.833	-0.378	1.000	17.503	1.000
Instab.	4	-6.634	0.000	0.492	1.000	-0.397	1.000	15.028	1.000

The stable subset has average posterior means $\bar{\beta}_1 = 0.493$ and $\bar{\beta}_2 = -0.378$. The numerical-instability subset has average posterior means $\bar{\beta}_1 = 0.492$ and $\bar{\beta}_2 = -0.397$. Since the truths are $\beta_1 = 0.5$ and $\beta_2 = -0.4$, the regression slopes are well recovered. The main weakness is latent-scale recovery and numerical stability.

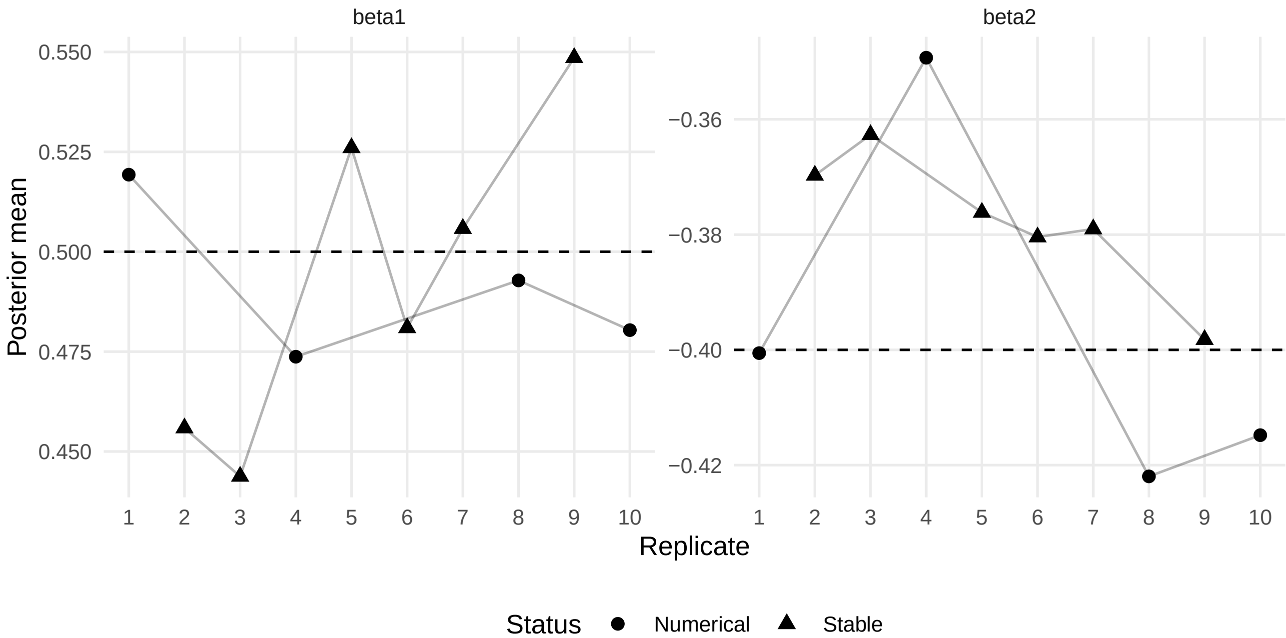

Figure 2: Posterior means of beta1 and beta2 by replicate. Dashed horizontal lines show true values.

7 Replicate-level posterior diagnostics

Table 6: Replicate-level regression and dispersion summaries.

Rep	Status	Mean count	Zero prop.	b0	b1	b2	r
1	Instab.	2.226	0.359	-7.776	0.519	-0.401	14.412
2	Stable	3.130	0.173	-5.712	0.456	-0.370	17.851
3	Stable	2.911	0.176	-5.768	0.444	-0.363	16.418
4	Instab.	1.958	0.312	-6.383	0.474	-0.349	16.041
5	Stable	4.364	0.173	-5.819	0.526	-0.376	16.821
6	Stable	2.893	0.233	-5.996	0.481	-0.380	16.826
7	Stable	2.976	0.309	-5.983	0.506	-0.379	18.304
8	Instab.	2.620	0.312	-6.404	0.493	-0.422	15.539
9	Stable	2.869	0.218	-5.882	0.549	-0.398	18.798

Rep	Status	Mean count	Zero prop.	b0	b1	b2	r
10	Instab.	3.108	0.223	-5.972	0.480	-0.415	14.120

Table 7: Replicate-level latent-risk recovery and numerical guards.

Rep	Status	lambda RMSE	log RMSE	cor(log)	cor(delta)	kappa guard	lambda guard	max log lambda
1	Instab.	22,901.430	3.858	0.075	0.019	3	800,224	16.061
2	Stable	0.191	0.175	0.380	0.047	0	0	1.675
3	Stable	0.181	0.180	0.126	0.040	0	0	1.886
4	Instab.	1.125	0.575	0.523	0.151	0	10,836	5.366
5	Stable	0.182	0.170	0.058	-0.033	0	0	1.889
6	Stable	0.272	0.208	0.262	0.023	0	0	3.258
7	Stable	0.286	0.257	0.416	0.053	0	0	2.925
8	Instab.	3.193	1.184	0.304	-0.012	0	76,431	7.022
9	Stable	0.209	0.194	0.132	0.048	0	0	2.287
10	Instab.	0.574	0.680	0.464	0.123	0	14,970	3.766

Replicate 1 dominates the all-replicate lambda RMSE because it has a very large lambda-output guard count and a large stored $\log \lambda$ maximum. Replicates 4, 8, and 10 also trigger lambda-output guards, but their lambda-scale instability is less severe.

8 Latent temporal-risk recovery

Table 8: Lambda RMSE by fit-status group.

Group	N	Mean	Median	Max
All	10	2,290.764	0.279	22,901.430
Stable	6	0.220	0.200	0.286
Instab.	4	5,726.581	2.159	22,901.430

Table 9: Log-lambda RMSE by fit-status group.

Group	N	Mean	Median	Max
All	10	0.748	0.233	3.858
Stable	6	0.197	0.187	0.257
Instab.	4	1.574	0.932	3.858

Table 10: Latent-risk correlation, coverage, and maximum stored log lambda.

Group	N	cor(log)	cor(delta log)	lambda 95% cover	max stored log lambda
All	10	0.274	0.046	0.983	16.061
Stable	6	0.229	0.030	0.994	3.258

Group	N	cor(log)	cor(delta log)	lambda 95% cover	max stored log lambda
Instab.	4	0.341	0.070	0.966	16.061

The stable subset has mean λ RMSE 0.220 and mean $\log\lambda$ RMSE 0.197. However, the mean correlation for $\log\lambda$ is only 0.229, and the mean correlation for increments in $\log\lambda$ is 0.030. Thus, even stable fits do not recover the latent path strongly under sparse observations.

The all-replicate mean λ RMSE is not a useful ordinary performance average because it is dominated by one unstable replicate. The median and stable-subset summaries are more interpretable for posterior recovery, while the unstable-subset summaries quantify the failure mode.

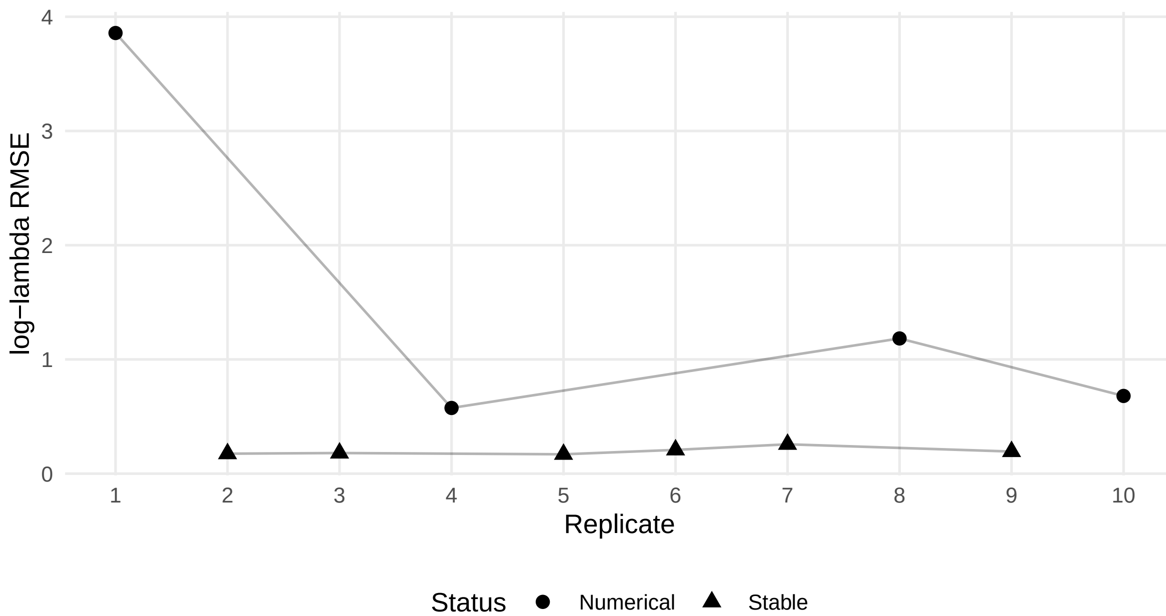

Figure 3: Log-lambda RMSE by replicate.

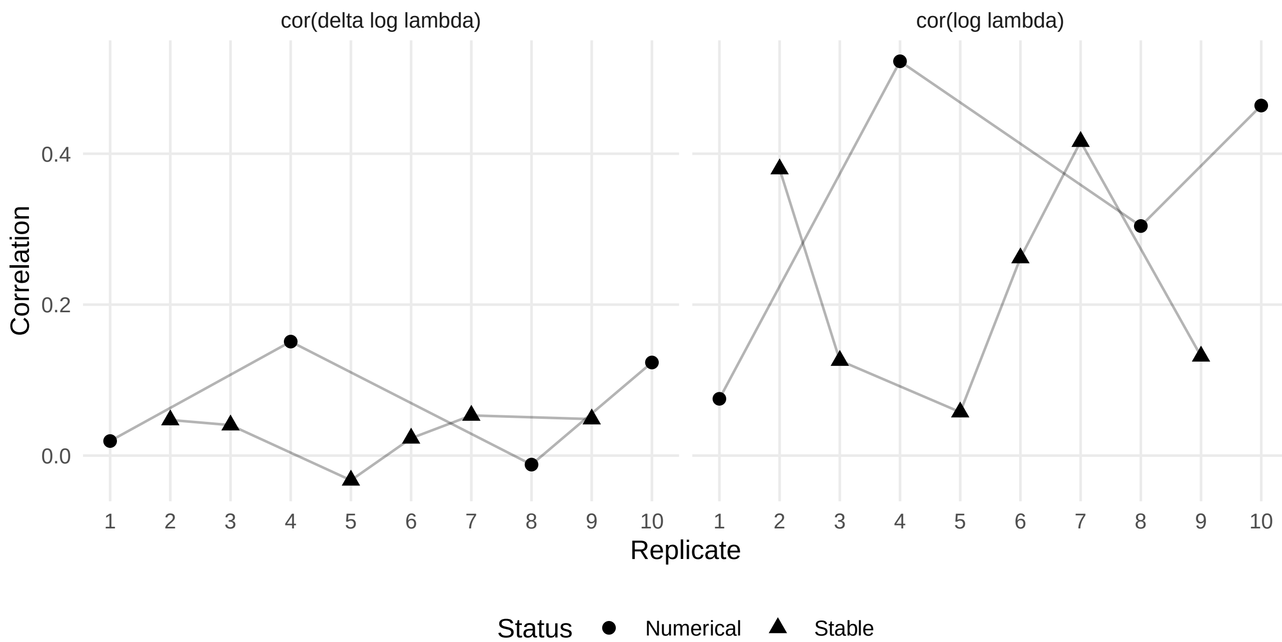

Figure 4: Correlations for log lambda and increments in log lambda by replicate.

9 Numerical guards

Table 11: Numerical guard summary by subset.

Subset	# reps	beta	kappa	lambda input	lambda output	lambda output rate
All	10	0	3	0	902,461	0.002507
Stable	6	0	0	0	0	0.000000
Instab.	4	0	3	0	902,461	0.006267

No beta guards occurred in the main fits. The numerical issue is concentrated in the lambda-output block. The total lambda-output guard count is 902,461, and most of it occurs in the numerical-instability subset.

10 Oracle diagnostics

The oracle diagnostics were run for the four numerical-instability replicates: 1, 4, 8, 10.

Two diagnostics were used:

1. **Oracle lambda:** fix λ at the truth and estimate β, ϕ, r, κ.
2. **Oracle lambda plus oracle phi:** fix both λ and ϕ at the truth and estimate β, r, κ.

These diagnostics ask whether the numerical failures are caused by the regression/dispersion update itself or by joint estimation of latent effects under sparse observations.

10.1 Oracle lambda

Table 12: Oracle-lambda regression and dispersion summaries.

Rep	Mean	Zero	b0 truth	b0	b1	b2	r
1	2.226	0.359	-6.385	-6.405	0.513	-0.403	13.516
4	1.958	0.312	-6.249	-6.175	0.482	-0.361	13.809
8	2.620	0.312	-6.047	-6.091	0.505	-0.421	14.162
10	3.108	0.223	-5.859	-5.843	0.486	-0.413	13.193

Table 13: Oracle-lambda latent-effect checks and guards.

Rep	phi RMSE	phi cor	lambda RMSE	log RMSE	beta guard	kappa guard
1	0.133	0.998	0.000	0.000	0	41,049
4	0.165	0.997	0.000	0.000	0	0
8	0.142	0.996	0.000	0.000	0	0
10	0.115	0.984	0.000	0.000	0	0

Fixing λ restores regression recovery in all problem replicates. In particular, replicate 1 changes from a main-fit intercept mean around -7.776 to an oracle-lambda intercept mean around -6.405, close to its identified truth -6.385. This confirms that the major failure in replicate 1 is a latent-scale issue rather than a slope-estimation issue.

Replicate 1 still has kappa guards under oracle lambda alone. This indicates that fixing λ is helpful but does not completely remove all latent-effect compensation.

10.2 Oracle lambda plus oracle phi

Table 14: Oracle-lambda-plus-phi regression and dispersion summaries.

Rep	Mean	Zero	b0 truth	b0	b1	b2	r
1	2.226	0.359	-6.385	-6.374	0.513	-0.403	13.626
4	1.958	0.312	-6.249	-6.224	0.480	-0.349	14.066
8	2.620	0.312	-6.047	-6.058	0.506	-0.422	14.257
10	3.108	0.223	-5.859	-5.839	0.492	-0.405	12.912

Table 15: Oracle-lambda-plus-phi latent-effect checks and guards.

Rep	phi RMSE	phi cor	lambda RMSE	log RMSE	beta guard	kappa guard
1	0.000	1.000	0.000	0.000	0	0
4	0.000	1.000	0.000	0.000	0	0
8	0.000	1.000	0.000	0.000	0	0
10	0.000	1.000	0.000	0.000	0	0

Fixing both λ and ϕ eliminates the numerical guards in all four problem replicates. The beta and dispersion block is therefore not the source of the S4A failure. The main failure is caused by joint latent-scale estimation under sparse observations.

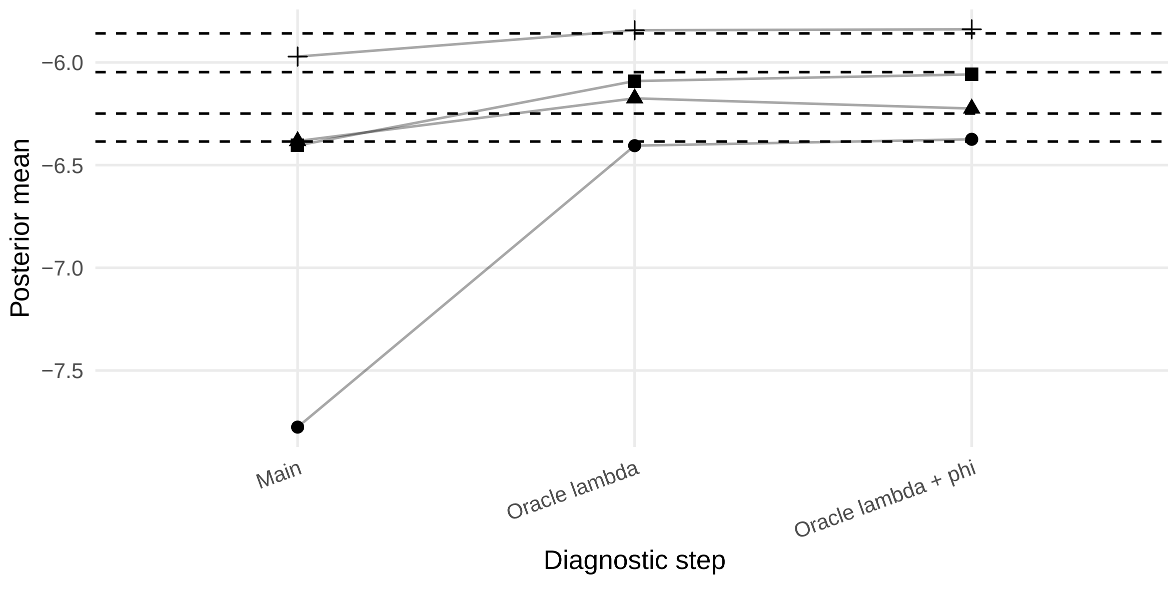

Figure 5: Beta0 recovery across main and oracle diagnostics for numerical-instability replicates.

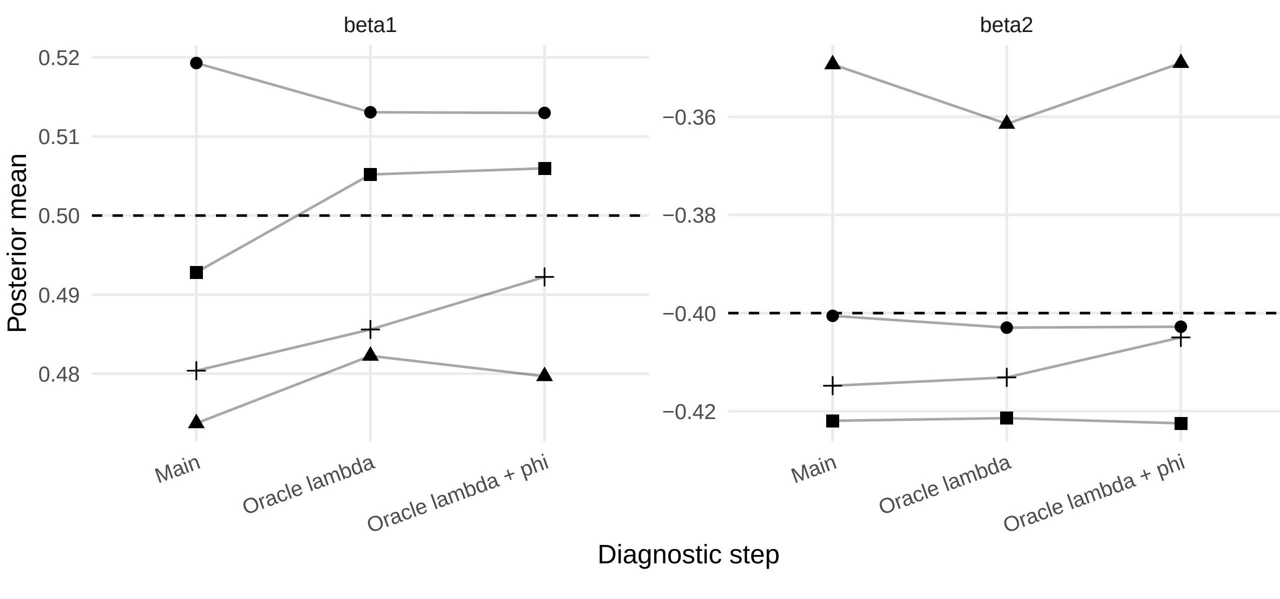

Figure 6: Slope recovery across main and oracle diagnostics for numerical-instability replicates.

11 Interpretation of the failure mode

The continuous-time x_2 S4A results separate two different phenomena.

First, regression slopes are recoverable. Both β_1 and β_2 are close to their true values in the all-replicate, stable, and numerical-instability subsets. This means the sparse-count stress does not destroy the slope information in this design.

Second, latent-scale recovery is fragile. Sparse observations weaken the information that separates the global intercept, spatial effects, and latent temporal risk. The sampled-λ fit can therefore move along a posterior ridge involving

$$\beta_0, \quad \phi_j, \quad \lambda_{tj}.$$

This ridge produces poor latent-risk recovery in selected replicates and triggers lambda-output numerical guards. Oracle diagnostics support this interpretation: fixing λ restores regression and spatial recovery, and fixing both λ and ϕ eliminates the remaining guards.

12 Practical implications

The S4A continuous-time x_2 results suggest the following practical implications for MSSTNB sparse-count applications.

1. Sparse observations should be treated as a latent-scale stress regime, even when regression slopes look accurate.
2. Fit-status classification is necessary. Averaging stable and numerical-instability replicates can hide the failure mechanism.
3. Stable-only summaries are useful for conditional recovery, but the numerical-instability rate is itself part of model performance.
4. High posterior coverage for λ does not necessarily imply strong path recovery; it may reflect wide intervals under weak information.
5. Oracle diagnostics are useful for separating implementation failures from practical identifiability failures.

13 Final conclusion

Scenario 4A with continuous-time x_2 confirms a practical sparse-count boundary of the MSSTNB model. The data have an average count of 2.905 and an average zero proportion of 0.249. Under this regime, 6 of 10 fitted replicates are stable and 4 show numerical instability.

The important distinction is that the regression slopes remain well recovered. The all-replicate posterior means are 0.493 for β_1 and -0.385 for β_2, compared with truths 0.5 and -0.4. The main degradation occurs in the latent temporal-risk process and the intercept-latent-scale relationship.

Oracle diagnostics show that fixing λ restores regression and spatial recovery in the problem replicates, and fixing both λ and ϕ removes numerical guards. Therefore, the S4A continuous-time x_2 failure mode is best interpreted as a sparse-count latent-scale identifiability problem, not as a regression-slope failure or a beta/r implementation failure.

14 Recommended reporting language

Scenario 4A evaluated the MSSTNB model under sparse observation-level counts while using the same continuous-time x_2 covariate design as the main dynamic simulations. The resulting data had an average count of approximately 2.91 and an average zero proportion of approximately 0.25. In the fixed-gamma sampled-λ fit, 6 of 10 replicates were stable and 4 triggered numerical instability. Unlike a slope-recovery failure, both β_1 and β_2 remained close to their true values in the stable and numerical-instability subsets. The main degradation occurred in the latent temporal-risk process and the intercept–latent-risk scale. Oracle diagnostics showed that fixing λ restored regression and spatial recovery, and fixing both λ and ϕ eliminated numerical guards. These results indicate that sparse observations create a practical latent-scale identifiability boundary for the full MSSTNB model.